

Class 5: Valid Argument Practice

Goal: Practice using basic rules of deduction to prove an argument is valid.

The homework exercises in the book provide a valid argument and all the steps used to prove the argument is valid. Your job is to give an explanation of how you know each step is valid.

Today we push this further so that you must come up with the steps and needed to prove the argument is valid.

Example 1: List the steps and provide justification to prove the validity of the argument.

$$(\sim A) \wedge (\sim B), (\sim C) \rightarrow B, C \rightarrow D \therefore D$$

Example 2: List the steps and provide justification to prove the validity of the argument.

$$(\sim A) \rightarrow B, (\sim C) \rightarrow (\sim A), (\sim B), C \rightarrow D \therefore D$$

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.

Example 3: List the steps and provide justification to prove the validity of the argument.

$$[(\sim A) \wedge B] \rightarrow C, \sim(A \vee C) \therefore \sim B$$

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.

Logical Quantifiers

Sentential logic as we have studied so far is not quite robust enough to handle more complicated mathematical statements. We need two more tools called logical quantifiers.

1. Existential quantifier:

2. Universal quantifier:

Next week we see how to prove (or disprove) such statements.